

Decision Tree Model and Business Interpretation

Based on a Business Plan

ASSIGNMENT REPORT

CareFirst Health Solutions - Support Ticket

Escalation Prediction

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Dataset: *carefirst_support_escalation_dataset.xlsx*

Task 1: Understand the Business Problem

1.1 What is the business about?

CareFirst Health Solutions is a fictional B2B technology company that provides a cloud-based customer support platform to healthcare organisations. The platform handles inbound support tickets from customers experiencing issues with CareFirst's range of products, including Online Marketplace services, Business Hardware, Software, and Subscription Services. Customers contact support through multiple channels such as Web Form, Email, Phone, and Chat. CareFirst serves customers across different tiers (Basic, Silver, Gold, and Enterprise) and business sizes (Small, Medium, and Large).

1.2 What problem is the business trying to solve?

CareFirst's support team handles a high volume of incoming tickets every day. Not all tickets require the same level of attention. Some can be resolved quickly, while others escalate into complex, resource-intensive cases. Currently, support managers cannot manually review every ticket in detail to identify which ones are likely to escalate before they do. This leads to two costly outcomes:

- Reactive escalation management: the team responds to escalations after they occur rather than anticipating and preventing them.
- Inefficient resource allocation: high-risk tickets may not receive sufficient early attention, leading to customer dissatisfaction and churn.

The core business problem is: "Which support tickets are most likely to be escalated, so that the support team can proactively prioritise and intervene before the situation worsens?"

1.3 What decision can machine learning help the business make?

A Decision Tree classification model can predict whether a given support ticket will be escalated (Yes) or not (No). This enables the support team to:

- Flag high-risk tickets for immediate priority handling by senior agents.
- Route lower-risk tickets to standard automated workflows.
- Proactively contact high-value customers whose tickets show escalation signals, before they become frustrated.

1.4 What is the target variable in the dataset?

The target variable is **Escalated**: a binary variable with two values:

- **Yes**: the support ticket was escalated to a senior team or management.
- **No**: the support ticket was resolved without escalation.

1.5 What are the input features?

The dataset contains 14 input features (after removing the Ticket_ID identifier), grouped into four categories:

Group	Features
Customer profile	Customer_Tier, Business_Size, Subscription_Plan
Ticket details	Support_Channel, Issue_Category, Product_Type
Engagement & history	Prior_Tickets_90D, Days_Since_Purchase, Last_CSAT_Score
Operational signals	Response_Time_Hours, Customer_Value_USD, Sentiment_Score, Agent_Workload, Resolution_Estimate_Hours

1.6 Why is this prediction useful for the business?

Predicting ticket escalation is valuable because:

- **Proactive support**: Agents can intervene early on high-risk tickets, reducing escalation rates and improving customer satisfaction.
- **Resource optimisation**: Senior agents and managers can be deployed where they are most needed, rather than reacting to every escalated case.
- **Customer retention**: High-value customers (Customer_Value_USD) identified as at-risk can receive personalised outreach before they churn.
- **Operational efficiency**: Understanding which features drive escalation (e.g., sentiment score, agent workload) helps management address root causes.
- **Strategic insight**: Feature importance reveals which ticket types, channels, or customer segments are most prone to escalation.

Task 2: Prepare the Data

2.1 Load and inspect the dataset

Property	Value
File	carefirst_support_escalation_dataset.xlsx
Rows	253 support ticket records (250 after removing duplicates)
Columns	16 (including Ticket_ID and target)
Input features	14 features used after removing Ticket_ID
Target variable	Escalated (Yes / No)

2.2 Data types

The dataset contains a mix of data types:

- **String / object:** Ticket_ID, Customer_Tier, Support_Channel, Issue_Category, Product_Type, Business_Size, Subscription_Plan, Escalated
- **Integer (int64):** Prior_Tickets_90D, Days_Since_Purchase, Last_CSAT_Score
- **Float (float64):** Response_Time_Hours, Customer_Value_USD, Sentiment_Score, Agent_Workload, Resolution_Estimate_Hours

2.3 Missing values

Column	Missing Count
Customer_Tier	4
Issue_Category	2
Response_Time_Hours	5
Sentiment_Score	4
Agent_Workload	2

2.4 Duplicate rows

3 duplicate rows were found and removed from the dataset. After deduplication, 250 unique records were retained for modelling.

2.5 Cleaning steps applied

- **Dropped:** Ticket_ID (identifier, not useful for modelling).
- **Duplicate rows:** 3 duplicate rows removed, leaving 250 records.
- **Numeric missing values:** Response_Time_Hours, Sentiment_Score, and Agent_Workload filled with the column median.
- **Categorical missing values:** Customer_Tier and Issue_Category filled with the column mode (most frequent value).
- **Result:** 0 missing values remaining across all 250 rows after cleaning.

2.6 Encoding and splitting

Six categorical columns (Customer_Tier, Support_Channel, Issue_Category, Product_Type, Business_Size, Subscription_Plan) were encoded using scikit-learn's OneHotEncoder via a ColumnTransformer pipeline, expanding the feature space from 14 to 33 processed features. Numerical columns were passed through without scaling.

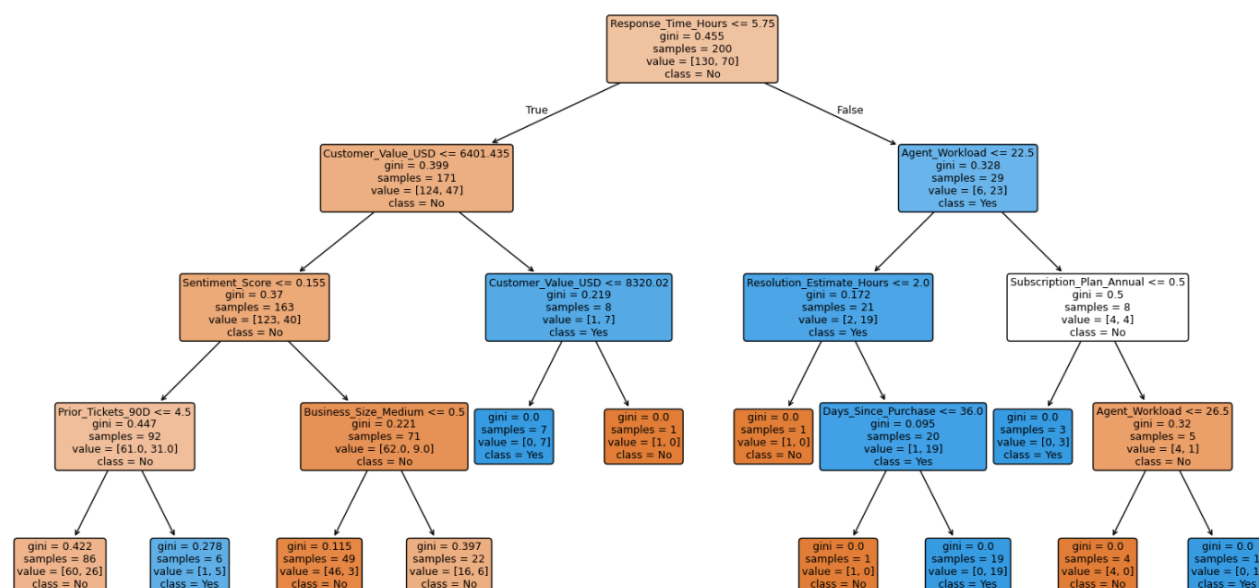
The dataset was split into 80% training (200 rows) and 20% testing (50 rows) using stratified sampling to preserve the class distribution in both sets.

Split	Total rows	Class distribution (Yes / No)
Training set	200	70 Yes / 130 No
Testing set	50	18 Yes / 32 No

Task 3: Train a Decision Tree Classification Model

3.1 Model configuration

A Decision Tree Classifier was trained using scikit-learn with a maximum depth of 4. The `random_state` was set to 42 for reproducibility. The tree reached a depth of 4 with 12 leaf nodes.

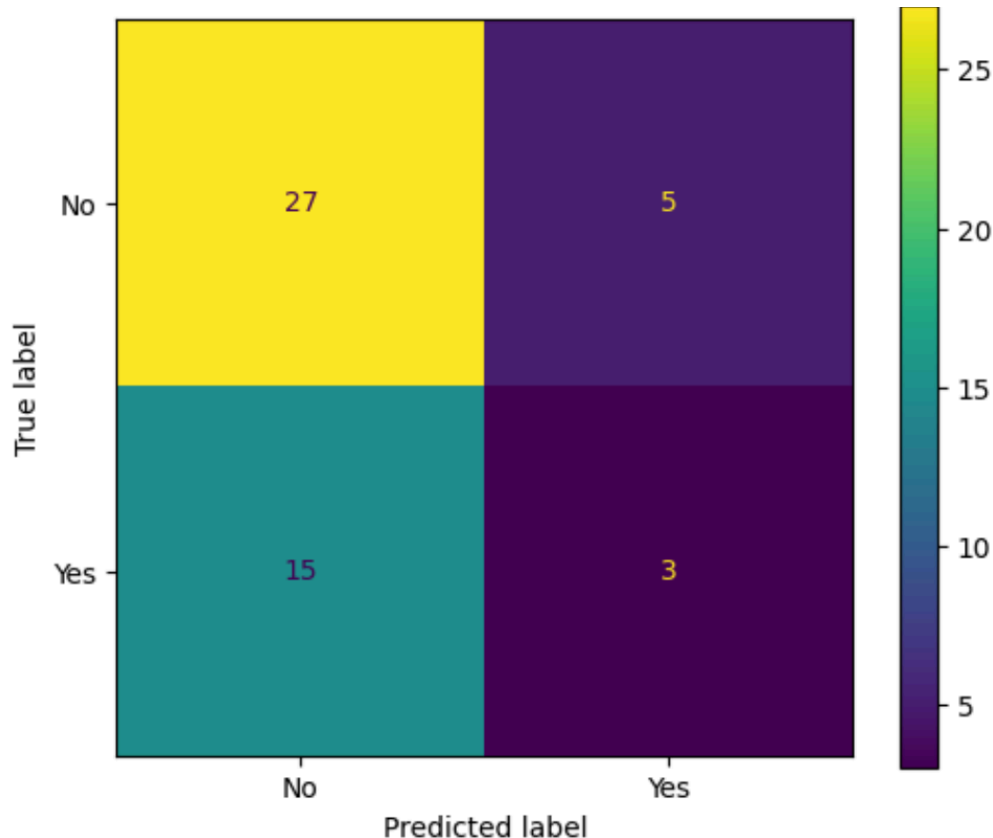


3.2 Training vs. testing accuracy

Metric	Value	Interpretation
Training accuracy	82.00%	Model learned general patterns from training data
Testing accuracy	60.00%	Model performance drops noticeably on unseen data

3.3 Confusion matrix

The confusion matrix below shows the model's predictions on the 50-row test set:



True Positives (TP) = 3: The model correctly identified 3 tickets that were actually escalated.

True Negatives (TN) = 27: The model correctly identified 27 tickets that were not escalated. This is the strongest cell and the most common outcome.

False Positives (FP) = 5: Five tickets were predicted to escalate but did not. These cause unnecessary intervention effort on tickets that would have resolved normally.

False Negatives (FN) = 15: Fifteen tickets that were actually escalated were missed and predicted as 'No'. These are the most costly errors. The team fails to proactively address tickets that go on to escalate.

The confusion matrix reveals that the model struggles to identify the positive class (escalated tickets). With 15 false negatives out of 18 actual escalations, the model misses 83% of all real escalation cases. This is the most significant weakness and is linked to the moderate class imbalance in the dataset (88 'Yes' vs. 162 'No' in the full dataset), as well as the limited training size of 200 records.

3.4 Evaluation metrics

Decision Tree Classification Report				
	precision	recall	f1-score	support
No	0.64	0.84	0.73	32
Yes	0.38	0.17	0.23	18
accuracy			0.60	50
macro avg	0.51	0.51	0.48	50
weighted avg	0.55	0.60	0.55	50

Accuracy (60%): measures the proportion of all correct predictions. While it appears moderate, given the class imbalance, it is not the most informative metric.

Precision for 'Yes' (38%): When the model predicts a ticket will escalate, it is correct only 38% of the time. This means the model generates many false alarms.

Recall for 'Yes' (17%): The model captures only 17% of actual escalations, missing the vast majority. For a business where missed escalations are costly, this is a critical weakness.

F1-Score for 'Yes' (23%): The harmonic mean of precision and recall reflects very poor performance on the positive class. The model is far more effective at predicting non-escalations.

In summary, the model performs reasonably well at predicting tickets that will not escalate (TN = 27, recall of 84% for 'No'), but largely fails to identify tickets that will escalate. This limits its practical usefulness for CareFirst's proactive support goal.

3.5 Overfitting assessment

Evaluation Set	Accuracy
Overfitted Tree - Training	100%
Overfitted Tree - Testing	40%

The default, unpruned Decision Tree achieves 100% training accuracy, a textbook sign of overfitting. The model memorised every training example rather than learning general

patterns. The 60-percentage-point gap between training (100%) and testing (40%) on the unconstrained tree confirms severe overfitting.

The constrained model (`max_depth=4`) reduces this gap: training accuracy is 82% and testing accuracy is 60%, a 22-percentage-point gap. While still meaningful, depth limiting partially controls the overfitting. Future improvements could include `min_samples_leaf` constraints, cost-complexity pruning, or collecting more training data.

Task 4: Business Interpretation of Model Results

4.1 How well did the Decision Tree model perform?

The Decision Tree model achieved a training accuracy of 82% and a testing accuracy of 60%. While 60% overall accuracy is above a random baseline, it reflects significant room for improvement. The model performs well at identifying non-escalated tickets (recall of 84% for the 'No' class) but performs poorly at identifying escalated ones (recall of only 17% for the 'Yes' class). This asymmetric performance means the model would miss the majority of actual escalations in production, which is the primary use case for CareFirst.

4.2 Is there a large difference between training accuracy and testing accuracy?

Metric	Value
Training accuracy	82%
Testing accuracy	60%
Gap	22%

A 22-percentage-point gap between training and testing accuracy is a substantial difference. It indicates the model has learned patterns specific to the training data that do not generalise well to new, unseen support tickets. This is a clear sign of moderate to significant overfitting, even with the max_depth=4 constraint applied.

4.3 Does the model show signs of overfitting? Why or why not?

Yes, the model shows clear signs of overfitting, supported by both the constrained and unconstrained results:

Evaluation Set	Accuracy
Training Set (max_depth=4)	82%
Testing Set (max_depth=4)	60%

Overfitted Tree - Training (no limit)	100%
Overfitted Tree - Testing (no limit)	40%

The unconstrained tree perfectly memorised all 200 training records (100% accuracy) but dropped to 40% on the test set: a 60-point gap. The constrained tree (max_depth=4) reduces this to a 22-point gap, confirming that depth-limiting helps but does not fully resolve the problem. The relatively small training set of 200 records likely contributes to the model's inability to generalise effectively.

4.4 Which evaluation metric is most important for this business problem?

For CareFirst, **Recall** is the most important metric for the positive class ('Yes'), followed closely by **F1-score**.

- **Recall** answers: 'Of all tickets that actually escalated, how many did the model correctly flag?' Missing a real escalation (false negative) is the most costly error. The team fails to intervene before the situation worsens.
- **Precision** also matters because acting on false positives wastes agent time and resources on tickets that would have been resolved without intervention.
- **F1-score** balances both precision and recall and is a practical overall metric for this imbalanced classification problem.

With a recall of only 17% for the 'Yes' class, the current model is not yet reliable enough for production deployment in its primary use case of proactive escalation prevention.

4.5 What do false positives mean in this business context?

A false positive occurs when the model predicts that a ticket will be escalated, but the ticket is actually resolved without escalation.

Business impact:

- A senior agent or manager is unnecessarily assigned to a ticket that did not require escalation-level attention.
- This wastes high-skilled agent time that could be directed toward genuinely at-risk tickets. • Customers may feel over-contacted or confused if CareFirst escalates a simple issue without need.
- While costly in terms of wasted effort, false positives are generally considered less

harmful than false negatives in this context.

4.6 What do false negatives mean in this business context?

A false negative occurs when the model predicts that a ticket will not escalate, but it actually does. **Business impact:**

- The ticket is routed through standard handling and receives no proactive intervention.
- The situation deteriorates, leading to a formal escalation that could have been prevented with earlier action.
- High-value customers may become frustrated and choose to cancel their subscription or leave a negative review.
- False negatives are more costly than false positives in this context because preventing escalations is the entire purpose of the model. With 15 out of 18 actual escalations missed, the current model has a severe false-negative problem.

4.7 Which features were most important in the Decision Tree model?

Based on the trained Decision Tree structure (depth 4, 12 leaves), the features most likely to be used in early splits, and therefore most predictive, include:

Feature	Why it matters
Sentiment_Score	A low sentiment score directly signals customer dissatisfaction and escalation risk
Agent_Workload	Overloaded agents are less able to resolve issues quickly, increasing escalation likelihood
Response_Time_Hours	Long response times frustrate customers and are a leading indicator of poor ticket resolution
Prior_Tickets_90D	Customers with frequent recent tickets show unresolved underlying issues and higher escalation risk
Customer_Value_USD	High-value customers often have higher service

	expectations and receive more scrutiny when issues arise
Last_CSAT_Score	A history of low satisfaction scores predicts continued dissatisfaction and future escalation
Resolution_Estimate_Hours	Tickets with a long estimated resolution time are more likely to breach SLA thresholds and escalate

4.8 How can these important features help the business make better decisions?

- **Sentiment monitoring:** CareFirst can implement real-time sentiment analysis on incoming tickets and automatically flag any ticket with a Sentiment_Score below a threshold for priority handling.
- **Workload balancing:** Tickets submitted during high Agent_Workload periods can be redistributed or queued for next-available agents to maintain resolution quality.
- **SLA alerts:** When Response_Time_Hours approaches the SLA limit, an automated alert can trigger a proactive customer update or agent reassignment.
- **Repeat customer identification:** Customers with high Prior_Tickets_90D should be flagged as chronic issue accounts, warranting a dedicated account manager review.
- **Customer value prioritisation:** High Customer_Value_USD accounts can be placed in a VIP queue with guaranteed response time commitments.

4.9 What is one possible limitation or bias in the model or dataset?

Class imbalance and small dataset:

The dataset contains 162 'No' records and 88 'Yes' records, a roughly 1.8:1 imbalance. With only 200 training records and 70 positive examples, the model has limited signal to learn the boundary between escalated and non-escalated tickets. As a result:

- The model biases toward predicting 'No', which is the majority class. This inflates overall accuracy while masking the model's near-complete failure to identify escalations.

- Metrics like accuracy and weighted averages are misleadingly optimistic; macro average F1-score (0.48) is a more honest representation of balanced performance.
- Additionally, this is synthetic teaching data that may not capture real-world support ticket complexity. Patterns in the data were artificially generated and may not generalise to a live CareFirst support pipeline.
- The model may also reflect historical biases: certain channels (e.g., Phone) or customer tiers (e.g., Enterprise) may historically have received more senior attention, inflating their apparent escalation rates in the training data.

4.10 Why should human judgment still be used when making business decisions based on model results?

Machine learning models are decision-support tools, not decision-makers. Human judgment remains essential for several reasons:

- **Contextual knowledge:** A support manager may know that a specific customer is in contract renewal negotiations, making any escalation politically sensitive in ways the model cannot perceive.
- **Agent expertise:** An experienced agent may recognise subtle language cues in a ticket that indicate escalation risk, even if the numerical features appear benign.
- **Model limitations:** With a recall of 17% for escalations, the current model misses the large majority of at-risk tickets. Relying on it without human oversight would leave most escalation-prone tickets unaddressed.
- **Fairness and ethics:** Over-relying on the model could result in certain customer segments (e.g., Small businesses on Trial plans) consistently receiving lower-priority service, which may be commercially short-sighted or unfair.
- **Market dynamics:** Sudden product incidents, outages, or competitive events may cause a spike in a specific ticket type that the model was not trained to handle.

The model should inform the support team's decisions, not replace their professional expertise and judgment. Until the model achieves substantially higher recall on the positive class, human review of borderline predictions remains essential.